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INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Computer Science & Engineering

End-Term Examination May – 2025

Course: B.Tech

Semester: - 4th

Branch: CSE

Subject Code: - 222

Max. Marks:60

Subject Name: Computer Organization and Architecture

Time: 03hr

NOTE:

1. All the questions are compulsory.
2. While solving write question number and its part clearly.
3. Solve questions in given sequence and all the sub parts should be at same place.
4. Answers without precision receive no credit.

1.A	Imagine you have thought to design your own computer architecture by using key components such as the CPU, memory, and I/O devices etc. How would you structure their interconnections and manage the flow of data within the system? Would you follow the Von Neumann model, the Harvard model, or create a hybrid of your own? Explain your design choices, and discuss the trade-offs involved in your approach.	5
B.	Consider 64MB main memory and 16KB cache divided into 128 bytes/block. Compute size of Physical address, no. of TAG bits, TAG comparators for the following mappings (Assume word size is one byte): (a) Direct mapping (b) Associate mapping (c) 8-way Set-associate mapping	5
2.A	Assume you are tasked with designing a hardware architecture for a cache memory mapping technique. To accomplish this, you are first required to analyze and perform a trade-off between different memory mapping techniques in terms of their hardware requirements and conflict issues. Based on your analysis, identify the mapping technique you prefer and design its hardware architecture. You may assume appropriate configurations for main memory and cache memory according to your design.	5
B.	Consider a disk having 256 tracks, 512 sectors per track, 2048 bytes/ sector. The disk is rotating with 3000rpm. Seek time is 10msec. Compute the followings: (a) Capacity of disk and number of bits required to address the disk. (b) It is required to read a file of 512 KB which was placed from 64th sector of outer track. The read-write head is currently position on the 4 th sector of Inner track. How much time is required to read this file.	5

3.A	A 32KB Direct mapped cache is organized into 128-byte blocks. A two-dimensional Array A 512x512 is manipulated by the program given below. Each element of the array occupies 8 bytes and array stored in row major order. What will be the total number of miss references with respect to the program <pre>for (i=0; i<512; i++) { for (j=0; j<512; j++) { X+=A[i] [j]; } }</pre>	5																									
B.	(a) If it is required to implement a 4-Stage Pipeline such that Stage-2 delay is twice the Stage-1 delay and half of Stage-3 delay. Stage-4 and Stage-2 delay are same. The Stage-3 delay is 40ns, then how much time is required to complete 100 instructions and what will be the speedup of this Pipeline? (b) Consider, a hypothetical pipeline in which different instructions spent different number of clocks. The following table use the number of clocks required for this non-linear pipeline. For each stage for each instruction, how many clocks are required to implement the following 4 instructions. <table><tr><td></td><td>S1</td><td>S2</td><td>S3</td><td>S4</td></tr><tr><td>I1</td><td>2</td><td>1</td><td>3</td><td>2</td></tr><tr><td>I2</td><td>1</td><td>1</td><td>2</td><td>3</td></tr><tr><td>I3</td><td>3</td><td>2</td><td>2</td><td>2</td></tr><tr><td>I4</td><td>1</td><td>1</td><td>1</td><td>1</td></tr></table>		S1	S2	S3	S4	I1	2	1	3	2	I2	1	1	2	3	I3	3	2	2	2	I4	1	1	1	1	3+2
	S1	S2	S3	S4																							
I1	2	1	3	2																							
I2	1	1	2	3																							
I3	3	2	2	2																							
I4	1	1	1	1																							
4.A.	Consider 4-stage instruction pipeline (IF, ID, Execute and Write back). Except the execution stage all the stages are taking 1 clock/instruction. Execution stage requires 3 clocks for multiplication and 1 clock each for addition and subtraction. What will be the minimum number of clocks required to compute following three instructions with and without the operand forwarding. Compare their speedup and performance also. I1 ADD R2 R1 R0 I2 MUL R4 R3 R2 I3 SUB R6 R5 R4	5																									
B.	Using an appropriate information flow diagram, explain the sequence of microoperations carried out during the Fetch, Execution, and Interrupt cycles in a basic computer system. Clearly distinguish the microoperations in each and describe how control signals coordinate these operations.	5																									

5.A	Consider a microprogramming control unit (CU), in which the system has to support 256-instructions. Each instruction on avg. requires $32\mu\text{-op}$. The system is using 63 control signals and 16 Flag conditions have to be supported by this CU. Further, one address control instruction is used for control word sequencing. Compute the space (in Kilobytes) required for control memory in case of Horizontal and Vertical microprogramming CU.	5
B.	With a neat schematic, discuss all the necessary steps involved in transferring data using Direct Memory Access (DMA) controller with proper instruction/signal flow . Also, compare the performance under its different operating modes (based on when the system bus is returned during the DMA process).	5
6.A	A process of 50MHz clock require 1500 clocks for initializing and terminating the DMA, it is required to transfer 400KB file from 2000KBps device. What is the percentage of processor time is used in this DMA.	5
B.	(a) The program below uses six temporary variables a, b, c, d, e, f. <pre> a = 1 b = 10 c = 20 d = a + b e = c + d f = c + e b = c + e e = b + f d = 5 + e return d + f </pre> Assuming that all operations take their operands from registers, what is the minimum number of registers needed to execute this program without spilling? (b) Analyze the number of 0-Address, 1-Address, 2-Address and 3-Address instruction that may be required to perform $X = (A - B) + (C * D) / E$	3+2